The Island Arc (1999) **8**, 426–439

Research Article

Paleomagnetism and tectonics of Karaginsky Island, Bering Sea

D. V. KOVALENKO[1] AND I. R. KRAVCHENKO-BEREZHNOY[2]

[1]*Institute of the Lithosphere, Russian Academy of Sciences, Staromonetnyi per. 22, Moscow, 109180 Russia (E-mail: dmitry@ilsan.msk.ru) and* [2]*Geological Institute, Russian Academy of Sciences, Pyzhevskii per. 7, Moscow, 109017 Russia*

Abstract Rocks from Karaginsky accretionary prism (Karaginsky Island, Bering Sea) yield both prefolding (close to original) and postfolding magnetic vectors. The prefolding vectors suggest that the Maastrichtian–Paleocene volcanic–terrigenous sequences of Karaginsky Island formed at approximately 40°N to 50°N ($n=45$, $D_G=325$, $I_G=57$, $K_G=6$, $\alpha_{95G}=8$, $F_G=15.06$, $D_S=332$, $I_S=63$, $K_S=20$, $\alpha_{95S}=4.5$, $F_S=0.3297$, $F_{cr}=2.64$) and were not originally part of either Eurasia ($F=19$, $\Delta F=6.5$) or North America ($F=17$, $\Delta F=4.4$). The geologic blocks rotated insignificantly counterclockwise about the horizontal plane, suggesting that the structure of Karaginsky Island arose without major strike-slip motions. Analysis of secondary magnetizations (for example, $n=28$, $D_G=311$, $I_G=-50$, $K_G=9$, $\alpha_{95G}=8.7$, $F_G=2.44$; $D_S=293$, $I_S=-41$, $K_S=5$, $\alpha_{95S}=11$, $F_S=12.04$, $F_{cr}=2.65$) reveals that the development of this framework involved at least two stages of deformation. During the second stage the sequences must have been tilted to west-northwest and northwest directions at 45–65°. This agrees with the northwest vergence of the structure of Karaginsky Island.

Key words: accretionary prism, Bering Sea, vectors of magnetization, deformations, paleolatitudes, vergence.

INTRODUCTION

Most reconstructions for the Bering Sea region consider Karaginsky Island to be part of an accretionary prism. This prism extends from the southern flank of the Malinovsky Range (Govena Peninsula) in the north via Karaginsky Island into the Eastern Ranges of Kamchatka (Zinkevich & Tsukanov 1992). The growth of this prism began by successive accretion of lithotectonic units of various ages in Early–Middle Eocene and still continues due to subduction of oceanic lithosphere (Bogdanov *et al.* 1982; Chekhovich *et al.* 1990; Kravchenko-Berezhnoy & Nazimova 1991).

The history of the Karaginsky accretionary prism is addressed by several geodynamic models. According to Kravchenko-Berezhnoy (1989), Kravchenko-Berezhnoy and Nazimova (1991) and Chekhovich *et al.* (1990) the convergence of plates was almost frontal during the entire period of the island structure formation. Another hypothesis (Giest *et al.* 1994) suggests that island arc blocks, which are located in the modern structure of the island, moved significantly sinistrally before the formation of the accretionary prism. During the accretionary prism formation the plate convergence direction varied and the strike-slip component presented over a period of time.

We have studied paleomagnetically lithotectonic units of different ages forming the Karaginsky accretionary prism in order to assess the rotation of the geological blocks forming the island structure, in the horizontal plane, to estimate whether these blocks were involved in the strike-slip movements during the different periods of the island structure formation.

GEOLOGIC FRAMEWORK OF KARAGINSKY ISLAND

On Karaginsky Island, several lithotectonic complexes are recognized (from oldest to youngest): ophiolitic, volcanic–siliceous, volcanic–terrigenous,

Accepted for publication 13 April 1999.
©1999 Blackwell Science Asia Pty Ltd.

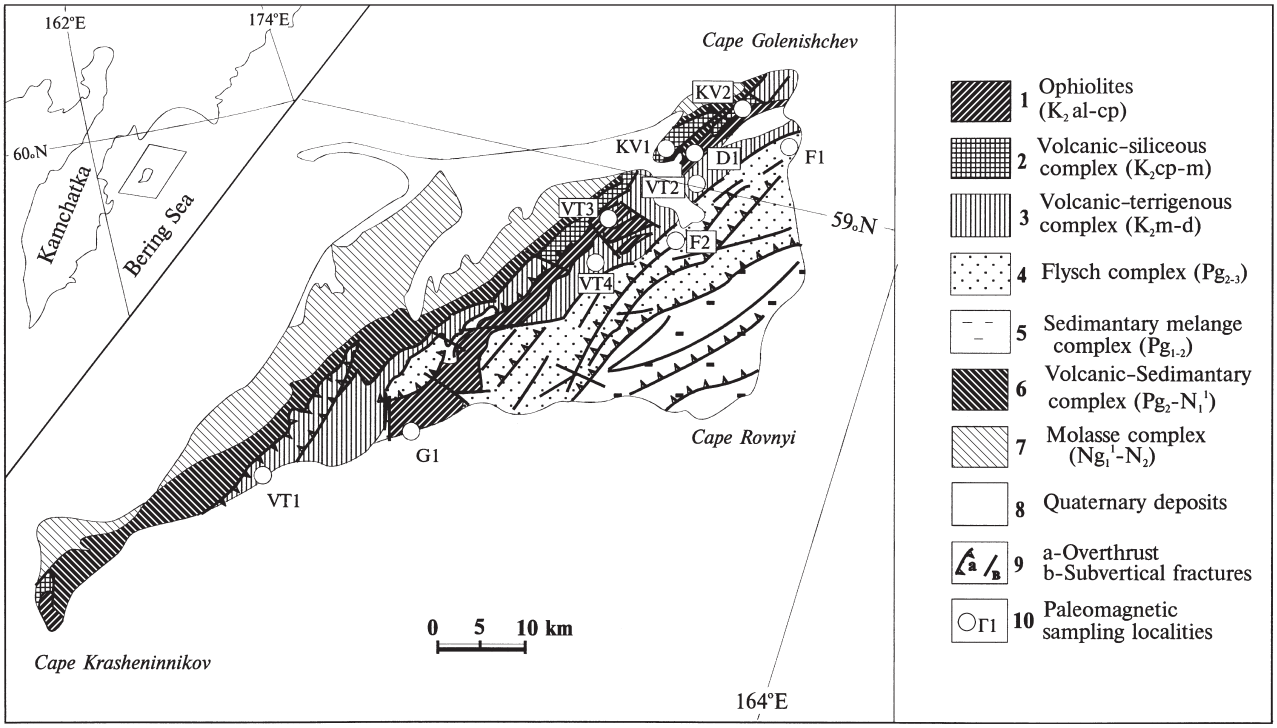


Fig. 1 Geologic scheme of Karaginsky Island (after Chekhovich *et al.* 1990).

sedimentary–volcanic, flysch, sedimentary melange, and molasse (Chekhovich *et al.* 1990) and Units 3, 2, 4, 1, 5 and 6 Kravchenko-Berezhnoy & Nazimova 1991. Distribution of these complexes are shown in Fig. 1.

The ophiolitic complex extends as a synform belt along the axis of the Central Range of the island and includes a serpentinite melange with slices and blocks of Alpine-type peridotites, both layered and massive gabbroids, sheeted dikes, and metamorphic rock (dynamothermal aureole). Petrologic and geochemical features reveal its supra–subduction zone affinity and suggest that the ophiolite complex originated in an island-arc or backarc-basin setting (Kravchenko-Berezhnoy 1989; Kravchenko-Berezhnoy & Nazimova 1991). K–Ar datings for the sheeted dikes yield values of 79–84 and 95–102 Ma (Kravchenko-Berezhnoy & Nazimova 1991), suggesting that they are of Albian–Campanian age.

The volcanic–siliceous complex is composed of tectonic blocks closely associated with the ophiolites. This complex consists of lava flows, lava breccias, tuffs, and cherts. The volcanics are calc-alkaline basalts to basaltic andesites, that formed in an ensimatic island arc (Kravchenko-Berezhnoy 1989; Chekhovich *et al.* 1990; Kravchenko-Berezhnoy & Nazimova 1991). These rocks are dated as Senonian by the presence of inocerams, and as Campanian–Maastrichtian by radiolarians (Chekhovich *et al.* 1990).

The volcanic–terrigenous complex forms a discontinuous stripe of fault-bounded blocks extending along the southeastern slope of the Central Range. The lower section of this complex (Khynkhlonai Formation), dominated by lavas and coarse to fine tuffs, and the upper section (Krutogorskaya Formation) conformably overlies the lower formation and consists of arc-derived terrigenous rocks—graywacke, siltstone, mudstone, sandstone, and tuffaceous varieties. The volcanics are calc-alkaline basalts to andesites formed in an evolved ensimatic island arc (Kravchenko-Berezhnoy 1989; Chekhovich *et al.* 1990; Kravchenko-Berezhnoy & Nazimova 1991). The lower section is dated as Maastrichtian–Danian, based on benthic foraminifers (Shapiro & Petrina 1985); K–Ar datings from basalts yield ages of 65–69 Ma, according to Melnikova *et al.* (personal communication, 1974). The upper section is dated as Danian–Paleocene based on foraminifers, as determined by Petrina from Melnikova's collections (personal communication, 1981).

The sedimentary–volcanic complex forms a northwest-dipping monocline that outcrops to the northwest of the aforementioned units and it is either overthrust or underthrust by them. The complex is composed of arc-derived volcaniclastics

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

that make up the Cape Tons Formation, Ilkhatun Formation, Laternula Sandstone Formation, and Pestrotsvetnaya Formation. Based on various fauna, including nannofossils, this complex is dated as Middle Eocene–Early Miocene (Gladenkov 1972; Chekhovich *et al.* 1990; Kravchenko-Berezhnoy & Nazimova 1991).

The flysch complex is located in the north-central part of the island. It includes the siltstone–sandstone member, flysch member, siltstone–spilite member, and siltstone member outlined by Melnikova *et al.* (personal communication, 1974). These members are flysch-like sequences with sporadic MORB-like basalt flow units and with occasional monotonous siltstone and mudstone members. Olistostrome horizons with fragments derived from the ophiolitic, volcanic–siliceous, and sedimentary–volcanic complexes are widespread. Radiolarian ages for these rocks are Middle Eocene. Nannofossil ages are latest Eocene–Early Oligocene (Kravchenko-Berezhnoy *et al.* 1990). Older rocks might be present as well. These deposits are interpreted to have formed in a deep-sea trench (Chekhovich *et al.* 1990) or a back-arc basin (Kravchenko-Berezhnoy *et al.* 1990).

The sedimentary melange complex outcrops in the southeastern part of the island, is underthrust beneath the flysch complex and comprises the siltstone–silicite Formation and Cape Rovnyi Formation. This complex consists of a sedimentary matrix with various inclusions. The matrix is composed of tectonized mudstone and siltsone and the inclusions are fragments, blocks and large bodies (possibly megaliths and olistoplates) of sandstone, chert, tuff and basalt. Some blocks contain conformable flow units of MORB-like tholeiites (Kravchenko-Berezhnoy 1989; Kravchenko-Berezhnoy *et al.* 1990). Various blocks contain planktonic foraminifers of Late Paleocene–Early Eocene age, radiolarians of Middle Eocene age (Chekhovich *et al.* 1990) and nannofossils of terminal Late Eocene–Early Oligocene age (from blocks with MORB-like basalts) (Kravchenko-Berezhnoy *et al.* 1990). This complex is thought to have formed in a deep-sea trench (Chekhovich *et al.* 1990) or, at least partially, in a back-arc basin (Kravchenko-Berezhnoy *et al.* 1990).

The molasse complex (terrigenous sequences of the Cape Ploskii, Yunyunvayam, Limimtevayam and Ust-Limimtevayam Formations) is located on the landward slopes of the island and overlies the older complexes with angular unconformity. Its stratigraphic range is Middle Miocene–Late Pliocene (Gladenkov 1972).

The structural relationship between these complexes is interpreted to be accretionary (Chekhovich *et al.* 1990). This relationship might be due to successive underthrusting of slices of the ophiolitic, volcanic–siliceous, volcanic–terrigenous, flysch and sedimentary melange complexes. The resulting thrust package was overlapped by the molasse complex. The accretionary structure verges to the southeast and may have been shaped by subduction-related processes that involved repeated deformation of the lithotectonic units (Bogdanov 1982; Kravchenko-Berezhnoy & Nazimova 1991).

PALEOMAGNETIC SAMPLING LOCALITIES

Samples for paleomagnetic study were collected from the ophiolitic, volcanic–siliceous, volcanic–terrigenous and flysch complexes.

The ophiolitic rocks were studied in two areas. Near Mount Shapochka (Fig. 1, locality G1), 11 samples were taken from a block of layered gabbro. On the north of the island eight samples were collected from diabase sheeted dikes (one sample per dike) in outcrops along one of the left tributaries of the Severnaya River (Fig. 1, D1).

The volcanic–siliceous complex was studied along the left tributaries of the Severnaya River (localities KV1 and KV2), where two sections, composed solely of pillow basalt flows, were sampled. Horizontal control of basalts was provided by the sporadically observed contacts of amygdular and massive basalts and by the voids (equally sporadic) in basalts, filled with layered mineral aggregates. We managed to collect a total of 18 samples from flow units with relatively reliable horizontal control.

The volcanic–terrigenous complex was sampled in four localities—in the south of the island (north of the Ilkhatunvayam River mouth; VT1 in Fig. 1) and in the north (near Mount Khynkhlonai (Fig. 1, VT2), in the basin of the Voronya River (Fig. 1, VT3), and at one of the tributaries of the Ketonitonvayam River (Fig. 1, VT4)). In the south, in sea cliffs to the north of the Ilkhatunvayam River mouth (locality VT1), samples were collected from a sequence of alternating fine tuffites and tuffaceous siltstones with syngenetic carbonate concretions. In places, this sequence is pierced by diorite dikes. The sequence is weakly deformed and is broken into six blocks with rock attitudes

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

which vary but insignificantly. We collected 83 samples from fine tuffs and from carbonate concretions.

The section at Mount Khynkhlonai (VT2) is dominated in its lower part by tuff members with rare siltstone intercalations. Upsection, the tuffs decrease in amount and black siltstones become dominant. This section contains benthic foraminifers (Shapiro & Petrina 1985) and from the lower part a single nannoplankton species was obtained (*Eiffellithus turriseiffelli*, as determined by Shcherbinina; personal communication, 1993). These findings suggest a Campanian–Maastrichtian age for this section. The sequence is broken into four blocks. A total of 56 samples from tuffites and tuffaceous siltstones were collected at roughly equal intervals from each block throughout the section.

In the basin of the Voronya River (VT3), samples were taken along three tributaries (Fig. 1). The upper parts of these sections are dominated by alternating black siltstones and mudstones; downsection, fine to coarse tuffites become dominant. From the siltstones, Shcherbinina determined two nannoplankton species—*Calculites obscurus* and *Cretarhabdulus crenulatus*—probably suggesting a Campanian–Maastrichtian age. Samples were collected from five blocks, bath from fine-grained tuffites and from black siltstones and mudstones.

In the basin of the Ketonitonvayam River (locality VT4), one small block composed of tuffaceous deposits was sampled.

The flysch complex was sampled in two blocks: in sea cliffs to the south of the Severnaya River (F1, Fig. 1) and on the Yaklekhrivayam River (tributary to the Markelovskaya River) (F2, Fig. 1). The F1 section is composed of black siltstones and mudstones with numerous carbonate concretions, and section F2 of black siltstones and mudstones lacking carbonate concretions. From section F1, Shcherbinina determined a single nannoplankton species (*Microrhabdulus decoratus*), characteristic of Cenomanian–Maastrichtian. Paleomagnetic samples (44 and 52, respectively) were collected from sedimentary rocks and concretions at roughly equal intervals along the sections.

LABORATORY TREATMENT OF PALEOMAGNETIC SAMPLES

From each hand sample, two cubes of 1 or 2 cm^3 were cut. Each cube was taken for thermal cleaning at 20–560°C, the upper limit being set by the appearance of newly formed magnetic phases that biased the remanent magnetization of the cubes. Most cubes were heated 8–10 times, the temperature being gradually increased by 50–60°C each time. After each step of the thermal cleaning vector directions (hereinafter designated as *J*nt) were measured on a JR-4 magnetometer. The thermal cleaning data from each cube were used to plot Zijderveld's diagrams (Zijderveld 1967) to perform component analysis of magnetization of the cubes (Kirschvink 1980) and for analysis of vector distribution over the sphere. For each sample, mean directions for the vectors isolated from all cubes were computed. A sample was rejected if it was impossible to compute the mean directions for any of the components with a confidence angle no greater than 25°.

All the components were conventionally grouped as hard and soft. Those magnetic components whose removal was accompanied by *J*nt vectors clearly moving along great circles were classified as soft. Accordingly, those components whose *J*nt vectors form end-points were termed hard. First, for each locality hard and soft components were analyzed separately (in multicomponent Zijderveld's plots, they are recorded jointly, yet they must have different origins). The directions of isolated components were then compared with the hard and soft components detected in single-component Zijderveld's plots. In case of coincidence of the areas occupied by hard and soft components isolated in single-component and multicomponent Zijderveld's plots, these hard or soft components were grouped into a single population.

All sequences were divided into blocks that comprised either fault-bounded fragments of sections or fragments of sections with distinctive rock attitudes. The directions of isolated components were analyzed in stereograms separately for each block and jointly for all the sampled blocks or groups of blocks (with similar rock attitudes) in the geographic and stratigraphic reference frames (Khramov *et al.* 1982).

RESULTS OF PALEOMAGNETIC STUDY

OPHIOLITIC COMPLEX

The NRM of samples from parallel dikes and layered gabbros yields only one component that persists to 500–550°C. In the gabbroic block this is a hard component with a reversed polarity and in

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

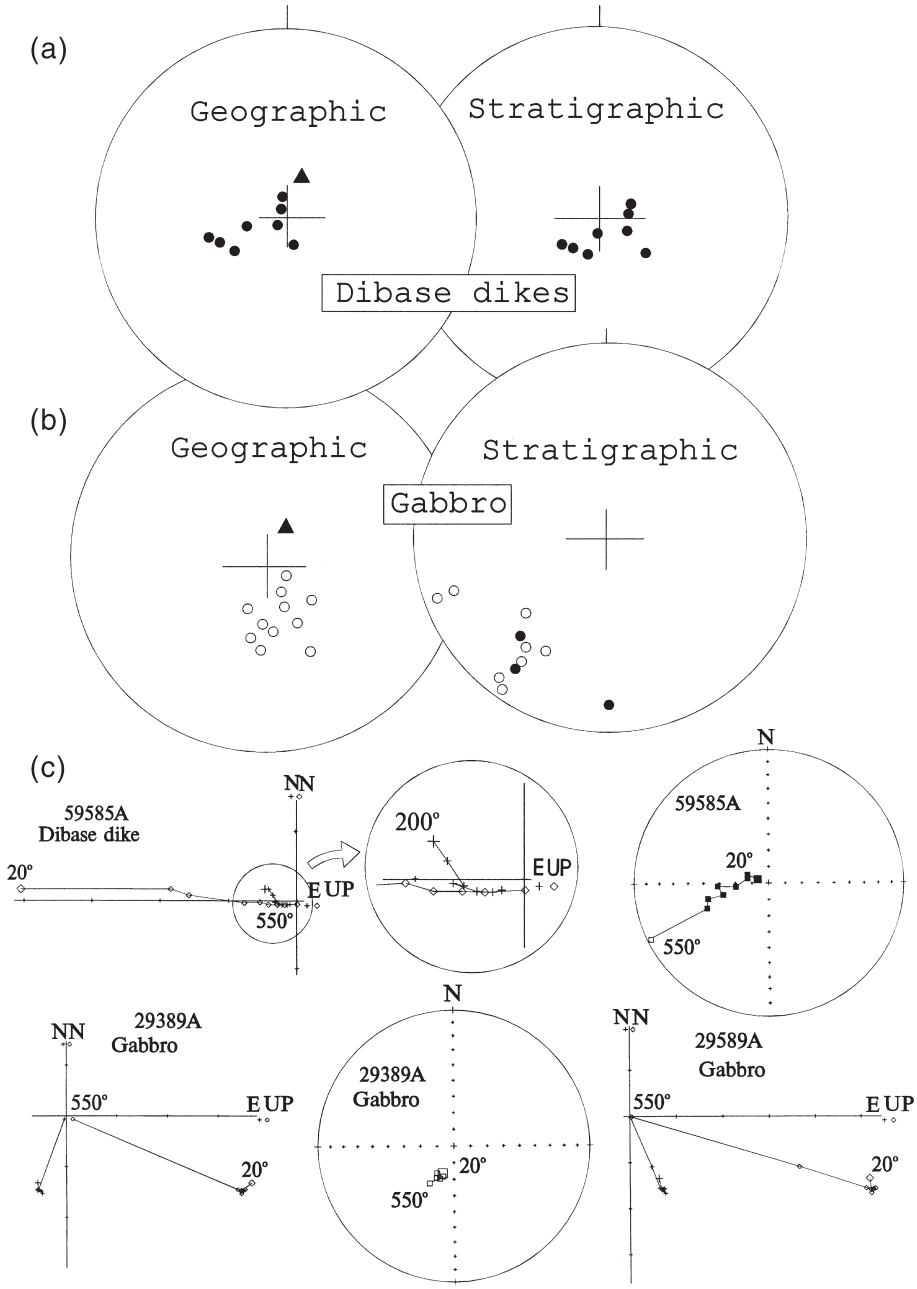


Fig. 2 Paleomagnetic characteristics of the ophiolite complex. (a) stereograms showing distribution of NRM vectors for sheeted dikes; (b) stereograms showing distribution of NRM vectors for layered gabbros; (c) typical Zijderveld's plots and *J*nt stereograms for the ophiolite complex in the geographic reference frame. ●, direct polarity; ○, reversed polarity; △, modern magnetic field for the study area. In Zijderveld plots, +, northeast, and ◇, Nup projection. ---, identified NRM components.

the dike complex it is a soft component with a normal polarity (Fig. 2). The *J*nt magnetic vectors of parallel dikes migrated during thermal cleaning along great circles. This suggests that the NRM of the dikes includes other, higher temperature components, which remained undetected during thermal cleaning. The *J*nt magnetic vectors of the gabbros did not change direction during thermal cleaning. The directions of components isolated are close (in the geographic reference frame) to the earth's magnetic dipole (normal and reversed) for this region. A fold test performed for the gabbroic block (Table 1) shows the vectors isolated from it to be of postfolding nature. No reliable tests could be applied to the parallel dikes.

VOLCANIC–SILICEOUS COMPLEX

The NRM of basalts from this complex chiefly yields a single magnetic vector. These vectors plot chaotically on the stereogram and are thus excluded from further consideration.

VOLCANIC–TERRIGENOUS COMPLEX

Rocks of this complex proved to have a rather complicated NRM. Zijderveld's plots commonly record

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1 Paleomagnetic data for the Karaginsky Island ophiolity

Locality	n/n_1	D_G	I_G	K_G	α_{95G}	D_S	I_S	K_S	α_{95S}
Mount Shapochka (G1), Severnaya River source, dikes (D1)									
(G1)	11/11	170	−62	36	7	223	−10	8	15
(D1)	8/8	253	74	25	10	151	81	25	10

n/n_1 number of NRM vectors put to analysis/number of samples collected from outcrops; D_G, I_G, K_G, α_{95G} declination, inclination, consistency, confidence angle in geographic reference frame; D_S, I_S, K_S, α_{95S} declination, inclination, consistency, confidence angle in stratigraphic reference frame. F_G, F_S, F_{Cr}: statistical criteria using, for comparison, average directions before and after tilt correction (McFadden & Jones 1981).

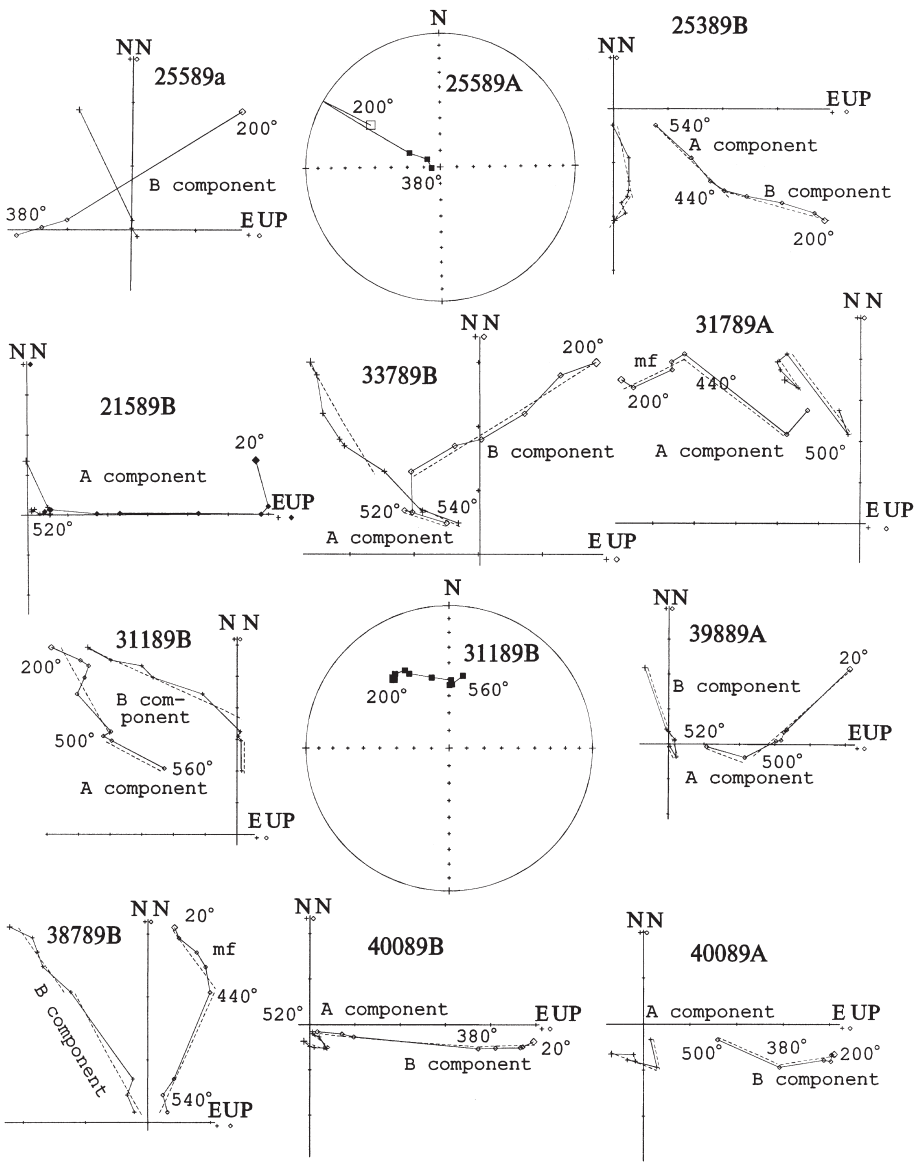


Fig. 3 Typical Zijderveld's plots and Jnt stereograms for the volcanic–terrigenous complex in the geographic reference frame. (For symbols, see Fig. 2).

either a single direction or several directions (Fig. 3). Soft components (e.g. Samples 25589A, 31189B in Fig. 3) are removed entirely in different samples by heating past 500–520°C. Hard components (A components in Fig. 3) persist to 540–560°C. Comparison of distribution of hard and soft components showed the soft components isolated in single- and multicomponent Zijderveld's plots to have similar directions; hence, they are further considered jointly. Some of them plot near the direction of the modern geomagnetic field (MF components) while others depart from this field drastically (B components). The area of hard directions computed from single-component plots includes both hard

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

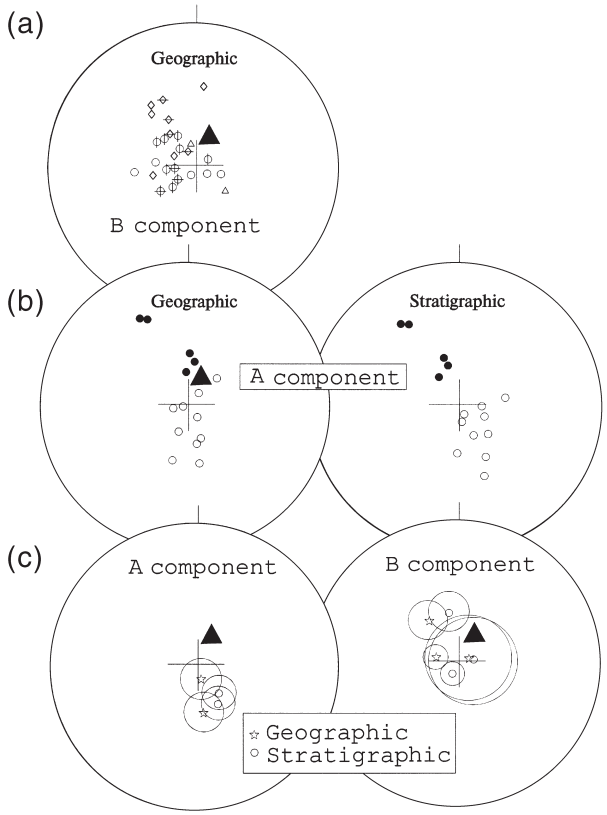


Fig. 4 Stereograms showing distribution of NRM components for locality VT1 (volcanic–terrigenous complex). (a) soft NRM components. (b) Hard NRM components. (c) Mean directions for soft and hard NRM components with confidence ovals. Different symbols correspond to NRM vectors of different geologic blocks. (For other symbols, see Fig. 2).

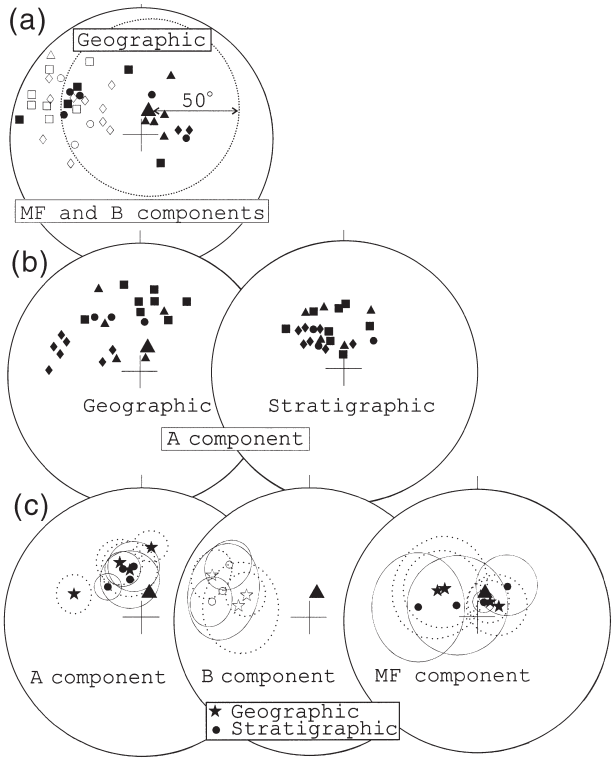


Fig. 5 Stereograms showing distribution of NRM components for locality VT2 (volcanic–terrigenous complex). (a) soft NRM components. Circle with a radius of 50° arbitrarily separates MF from B components. (b) Hard NRM components. (c) Mean directions for identified NRM components with confidence circles. Different symbols correspond to NRM vectors of different geologic blocks.

components obtained from multicomponent plots (A components) and, not infrequently, B components. Hence, the single-direction NRM of these samples might correspond either to the A component or to the later B component. Because they are commonly indistinguishable, all the samples with hard single-component NRM were discarded and further analysis dealt only with the hard A components isolated in Zijderveld's plots together with soft B components.

Samples from the Ilkhatunvayam River locality (VT1) yield only A and B components (Fig. 4, Table 2) computed from tuffites and carbonate concretions. The B components have a reversed polarity (Fig. 4a), and the A components have both normal and reversed pelarities (Fig. 4b). The fold test (comparison of means (McFadden & Jones 1981; Bazhenov & Shipunov 1988)), and reversal test performed for the A component confirm the hypothesis of its prefolding origin (Fig. 4c, Table 2). Averaged for blocks, the B component directions are not equal in either geographic or stratigraphic reference frames (Fig. 4c, Table 2), but the fact that they are nearly antipodal to the modern geomagnetic field of the study area and also to the remagnetization direction for the ophiolites suggests a postfolding origin for the B components.

Samples from Mount Khynkhlonai locality (VT2) yielded three distinct NRM components—MF, B and A (Fig. 5, Table 2). We assigned all the soft vectors that differ from the current geomagnetic field by less than 50° to the MF components, and all the remaining soft vectors to the B components (Fig. 5a). The distribution of A components with direct polarity is illustrated in Fig. 5b. Fold tests showed that the A components predate folding and block means of the B and MF directions are equal in both geographic and stratigraphic reference frames (Fig. 5c).

Samples from the Voronya River basin (VT3) also yielded all three components—A, B and MF (Fig. 6, Table 2). Fold tests revealed that the A components predate folding and B components postdate folding (Fig. 6c, Table 2).

In the Ketonitonvayam River locality (VT4), only A components were isolated (Fig. 7).

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2 Paleomagnetic data for volcanic–terrigenous complex of Karaginsky Island

Components	n/n_1	D_G	I_G	K_G	α_{95G}	D_S	I_S	K_S	α_{95S}
Locality I (Ilkhatunvayam River mouth) (VT1)									
AR	10	169	−77	21	10	146	−70	27	8.5
AN	5	348	55	15	16	333	54	18	15
	$F_{2g}=0.3317$		$F_{2s}=0.23$				$F_{2kr}=0.2591$		
AI,III,V	7	160	−80	20	12	144	−69	23	11
AIV,VI	8	351	60	18	12	332	61	19	11
	$F_{2g}=0.3201$		$F_{2s}=0.065$				$F_{2kr}=0.259$		
AΣ	15/83	348	70	15	9	329	65	21	8
BI,III,V	15	278	−80	26	7	194	−81	25	7
BII	2	88	−82	14	26	107	−80	14	26
BIV,VI	9	328	−63	19	11	349	−63	19	11
BΣ	26/83	311	−77	16	7	328	−86	12	8
	$F_g=5.69$		$F_s=11.47$				$F_{(4,46,95)kr}=2.64$		
Locality II (Mount Khynkhlonai) (VT2)									
AI	8	7	45	18	12	341	59	19	11
AII	3	340	53	30	15	348	67	16	20
AIII	7	291	46	23	11	316	61	44	8
AIV	5	348	58	10	20	348	57	12	18
AΣ	23/56	336	54	8.5	10	335	61	19	6.6
	$F_g=8.01$		$F_s=1.14$				$F_{(6,38,95)kr}=2.67$		
AIrot	8	350	−3	18	12	354	59	18	11
AIIrot	3	331	−5	30	15	348	67	18	19
AIIIrot	7	299	−11	23	11	301	61	40	8
AIVrot	5	333	2	10	20	348	57	13	17
AΣrot	23					336	62	16	7
			$F_s=2.9$				$F_{(6,38,95)kr}=2.67$		
BI	8	296	−20	15	13	306	−28	11	14
BII	3	283	−43	9	27	290	−32	9	27
BIII	10	295	−46	11	13	284	−25	12	13
BIV	1	315	−8			316		1	
BΣI,II,III	21/56	294	−36	10	9.6	293	−28	11	9.3
	$F_g=1.44$		$F_s=1.39$				$F_{(6,34,95)kr}=2.7$		
CΠI	4	314	59	5	32	288	53	5	32
CΠII	5	318	60	5	27	313	72	4	31
CΠIII	2	54	72	23	20	43	60	31	18
CΠIV	5	38	72	30	11	35	72	159	5
CΠΣ	16/56	344	70	7	13	339	74	6	14
	$F_g=1.5618$		$F_s=2.0412$				$F_{(6,24,95)kr}=2.77$		
Locality III (Voronya River basin) (VT3)									
AII	7	173	−84	55	7	271	−63	45	8
AIII	5	88	−73	42	10	214	−81	31	11
AIV	5	130	−72	19	14	267	−72	20	14
AV	5	267	−65	14	17	282	−62	11	19
AΣ	22/99	159	−85	14	8	268	−70	18	7
	$F_g=7.09$		$F_s=2.3049$				$F_{(6,36,95)kr}=2.67$		
AIIrot	7	148	−30	55	7	262	−63	41	8.5
AIIIrot	5	130	−25	42	10	199	−81	31	11
BI	7	324	−42	11	16	320	−74	17	13
BII	11	314	−45	10	13	305	−19	10	13
BIII	10	292	−60	10	14	271	−37	8	16
BΣ	28/99	311	−50	9	8.7	293	−41	5	11
	$F_g=2.447$		$F_s=12.07$				$F_{(4,50,95)kr}=2.65$		
CII_I	8/99	318	71	9	17	316	34	6	21
Locality IV (Ketonitonvayan River basin) (VT4)									
A	7/7	282	19	12	15	329	64	15	14

For designations, see Table 1.

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

FLYSCH COMPLEX

Meaningful results were obtained only at locality F1. Samples of black siltstones and carbonate concretions collected from two blocks at this locality yielded only soft components (Fig. 8, Table 3). Fold tests suggest a postfolding origin for the NRM of the flysch sequences.

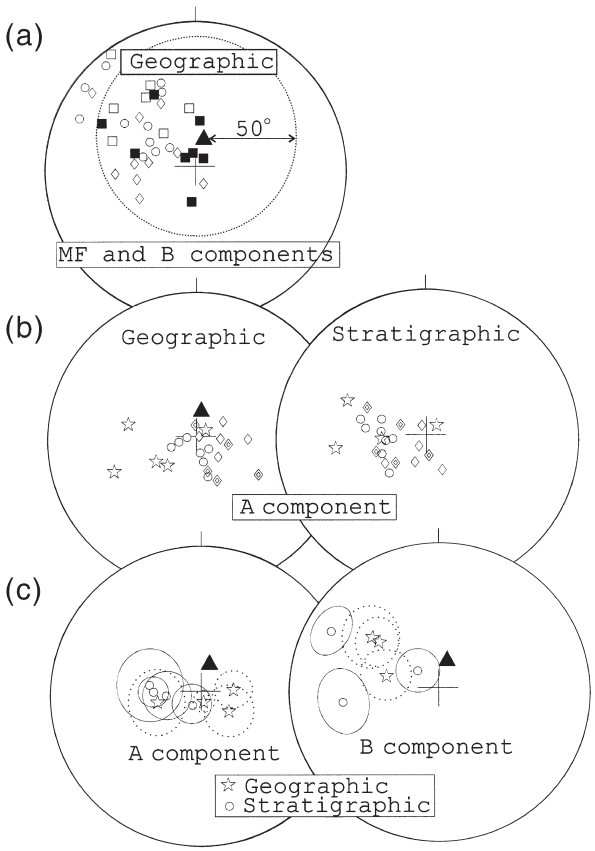


Fig. 6 Stereograms showing distribution of NRM components for locality VT3 (volcanic–terrigenous complex). (a) Soft NRM components. Circle with a radius of 50° arbitrarily separates MF from B components. (b) Hard NRM components. (c) Mean directions for identified NRM components with confidence circles.

ANALYZING VECTORS OBTAINED

As shown above, rocks from various lithotectonic units of Karaginsky Island yielded several distinctly directed NRM vectors.

Group 1 of the vectors includes soft components coinciding with the present geomagnetic field in the geographic reference frame. These are the MF components obtained from the volcanic–terrigenous complex and NRM components from the parallel dikes of the ophiolitic complex. Apparently, these components are the youngest because they are close to the modern geomagnetic field of the region and the fold test performed for all the MF components of the volcanic–terrigenous complex (localities VT2, VT3) shows that the MF components postdate folding ($n=24$; in geographic reference frame: $D=335$, $I=72$, $K=8$, $\alpha_{95}=10$, $F_G=2.16$; in stratigraphic reference frame: $D=325$, $I=61$, $K=5$, $\alpha_{95}=13$, $F_S=3.29$, $F_{cr}=2.67$).

Group 2 comprises secondary reversed-polarity magnetizations that are approximately antipodal to the geomagnetic field for Karaginsky Island in the geographic reference frame. We attribute NRM components from the ophiolitic layered

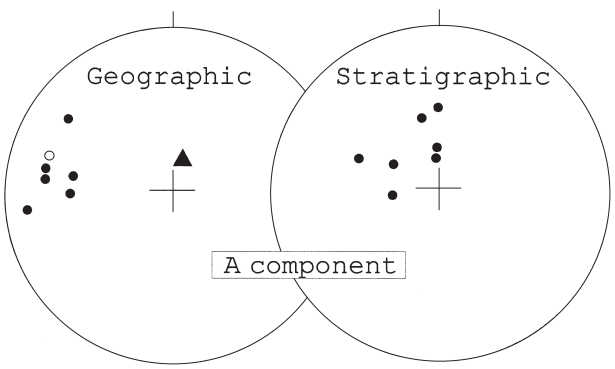


Fig. 7 Stereograms showing distribution of NRM components for locality VT4 (volcanic–terrigenous complex). (For symbols, see Fig. 2.)

Table 3 Paleomagnetic data for flysch complex of Karaginsky Island

Components	n/n_1	D_G	I_G	K_G	α_{95G}	D_S	I_S	K_S	α_{95S}
Severnaya River mouth (F1)									
BI	7/7	295	−26	20	12	295	28	18	12
BIIsed	15/37	310	−15	8	13	311	12	7	14
BIIcon	19/37	298	−42	11	9.5	324	−16	10	10
BII	34/37	304	−31	8	9	318	−4	7	9
B∑I,II	41/44	302	−30	9	7	314	2	6	9
	$F_g=0.0164$		$F_s=0.24$				$F_{2kr}=0.08$		

Sed, sedimentary rocks; con, concretions; for other designations see Table 1.

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

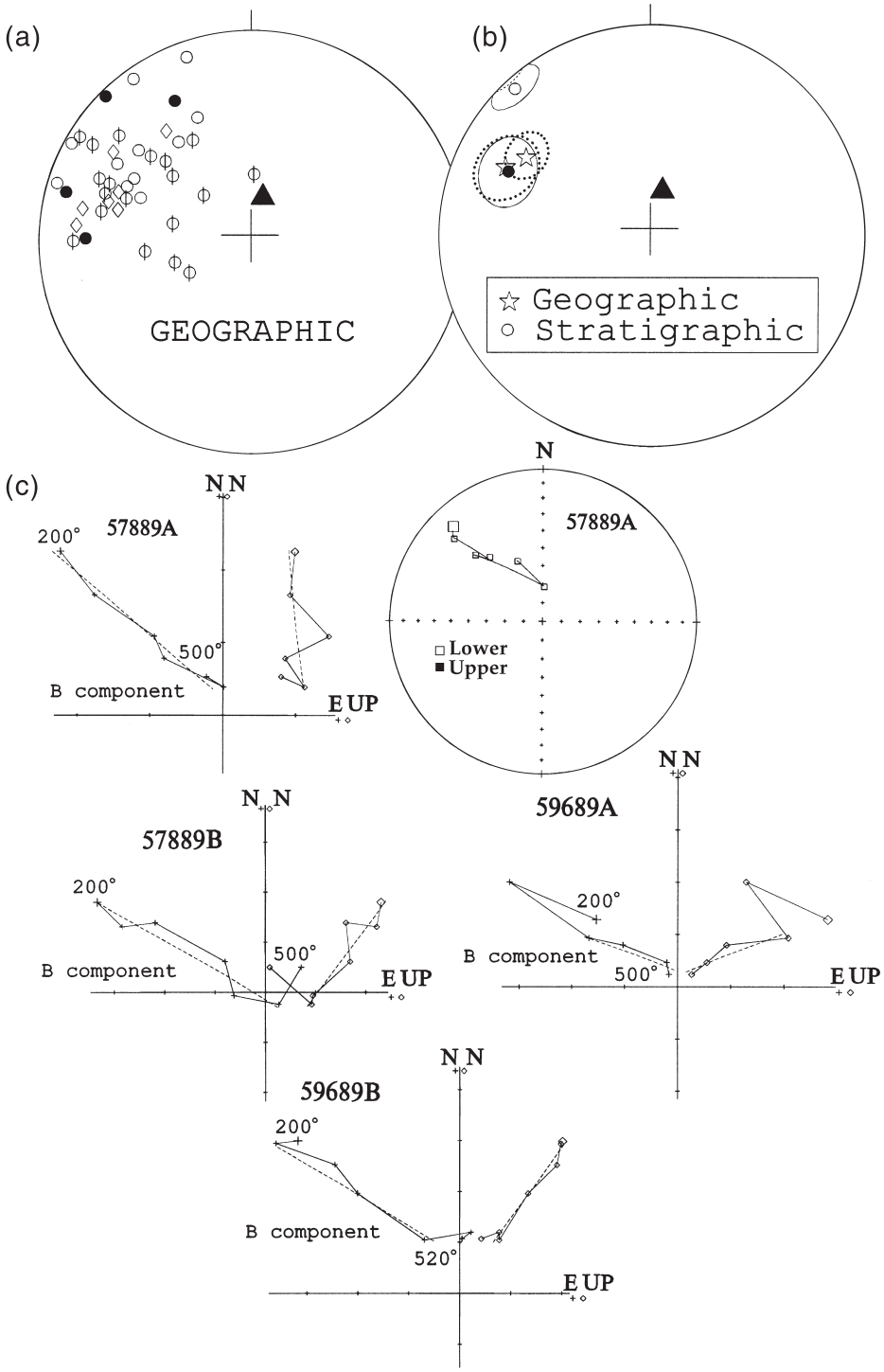


Fig. 8 Paleomagnetic characteristics for the flysch complex. (a) stereograms showing distribution of identified NRM components. ◇, block I; ●, block II (crossed, concretions, and open, sediments). (b) stereogram showing distribution of mean NRM components with confidence ovals. (c) typical Zijderveld's plots and *J*nt stereograms in the geographic reference frame. (For symbols, see Fig. 2.)

gabbros and the B components isolated from the VT1 sequence of the volcanic–terrigenous complex to this group. Their origin is assumed to be secondary based on the fact that the gabbroic rocks failed the fold test, and also on the coexistence in the VT1 locality of soft B components and hard (apparently primary, judging from fold and reversal tests) A components.

Group 3 (probably also secondary) comprises B components that depart drastically from the local geomagnetic field, both direct and reversed. These are B components from localities VT2, VT3 and F1. Their secondary origin is proved by the failed fold tests for localities VT3 and F1. Because these stable (in the geographic reference frame) components are isolated from sections with different rock

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

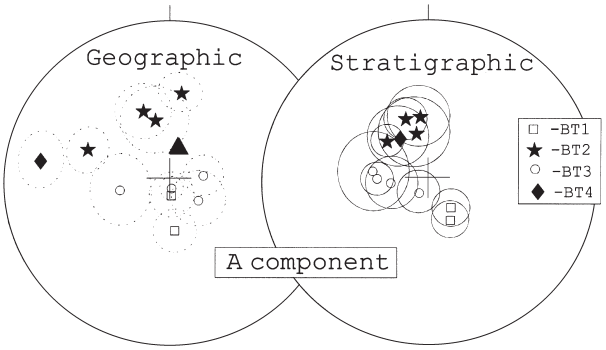


Fig. 9 Stereograms showing distribution of mean A components for all the localities of the volcanic–terrigenous complex.

attitudes (in F1 blocks, rocks dip 45° at an azimuth of 185°; in VT3 blocks, rocks dip 30° at azimuths of 320, 100 and 60°; in VT2 blocks, rocks dip 25° at an azimuth of 230°, 15° at an azimuth of 150°, and 30° at an azimuth of 65°) and because they coexist with A vectors of both normal and reversed polarities, these components must be independent, rather than the result of summing different but equally stable components (e.g. A and secondary components of the first two groups).

Lastly, Group 4 includes hard A components isolated from the rocks of the volcanic–terrigenous complex. Their mean directions for all the sampling localities (VT1, VT2, VT3, VT4) are shown in Fig. 9. Rocks from locality VT1 passed both fold and reversal tests and rocks from localities VT2 and VT3 passed fold tests. The regional fold test is positive for localities VT1, VT2 and VT4 ($n=45$, $D_G=325$, $I_G=57$, $K_G=6$, $\alpha_{95G}=8$, $F_G=15.06$; $D_S=332$, $I_S=63$, $K_S=20$, $\alpha_{95S}=4.5$, $F_S=0.3297$, $F_{cr}=2.64$). The A components for locality VT3 differ from the other localities, possibly because the blocks at this locality rotated in the horizontal relative to VT1, VT2 and VT4 blocks. However, the fact that the A vector of sequence VT3 is antipodal to the present geomagnetic field and to Group 2 postfolding components as well as the structural setting of the VT3 sections among remagnetized ophiolitic rocks (Fig. 1) suggest a secondary origin for the A component of the VT3 volcanic sequences. These sequences may have acquired a new magnetic overprint as a monoclinally dipping structure to be then deformed once more.

TECTONIC IMPLICATIONS OF PALEOMAGNETIC DATA

From a tectonic standpoint, only the B vectors of Group 3 and the A vectors are of interest.

The A vectors (except those from locality VT3) indicate that the Maastrichtian–Paleocene island-arc sequences on Karaginsky Island formed at about 40–55°N and are thus probably fragments of either the East Kamchatka island arc or the Olyutorsky island arc. Deposits of these arcs are found in the Olyutorsky block (Chekhovich *et al.* 1990) and in the Kamchatka Isthmus.

Estimates based on Beck's procedure (Beck 1980; Demarest 1983) using reference paleomagnetic poles for Eurasia and North America (Besse & Courtillot 1991) show that these sequences were not originally integratal to either Eurasia ($F=19$, $\Delta F=6.5$) or North America ($F=17$, $\Delta F=4.4$).

The studied blocks rotated slightly by $280\pm4.4°$ in a counterclockwise sense in the horizontal relative to the poleward direction, they rotated insignificantly ($R=-15$, $\Delta R=40$) relative to Eurasia, and by 440 ± 170 ($R=44$, $\Delta R=17$) in a counterclockwise sense with respect to North America. Such rotations are unlikely to be related to a strike-slip displacement in a subduction zone beneath the Karaginsky accretionary prism because similar values and senses of rotation have been obtained elsewhere in the northeastern Olyutorsky zone (Kovalenko 1992a,b; Kovalenko 1996) and in East Kamchatka (Levashova *et al.* 1999). Apparently, the insignificant rotations are related to more global processes that involved rotation of regional-scale blocks, rather than with processes that shaped the Karaginsky accretionary prism. Or, these rotations record the departure of declinations from the meridian, resulting in translational movement of these blocks as part of some plate over the earth's spherical surface.

As was noted above, the B component of Group 3 formed after a certain stage of deformation of the Karaginsky sequences. Because its direction does not coincide with the geomagnetic field for Karaginsky Island, which reset the NRM of the rocks, this direction must have departed from the remagnetizing field due to a new deformation increment. Hence, the paleomagnetic data suggest two stages of deformation for the sequences on Karaginsky Island, which is supported by the geologic evidence cited above.

The discrepancy between the B components and the direction antipodal to the modern geomagnetic field in the geographic reference frame poses constraints on the direction and magnitude of these deformations. The simplest model involves tilting of the sequences to the west-northwest and northwest directions by 45–65°. Such directions

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

conform with the northwesterly vergence of the Karaginsky structure. The fact that the A components show almost no horizontal rotation makes this model the most realistic. Because the mean B components for the blocks are equal in the geographic reference frame, the studied sequences of the volcanic–terrigenous and flysch complexes may have rotated as a single rigid slab, without significant internal deformations. Tilting these structures back in an east-southeast direction to align their B component with the remagnetizing geomagnetic field reveals that, prior to deformation (which is recorded paleomagnetically), the sequences dipped steeply (50–80°) to the southwest. The directions of A components in the stratigraphic reference frame based on rock attitudes for that position remained virtually unchanged (A_{rot} in Tables 2, 3), which further confirms the hypothesis that the framework of Karaginsky Island arose without significant rotations of its blocks.

The corollary that the oldest sequences in the Karaginsky accretionary prism experienced no rotation in the horizontal plane implies that plate convergence at this prism during its formation did not involve any significant strike-slip component, i.e. this convergence must have been nearly frontal. The accretionary prism formed in several stages through successive accretion of sequences of different ages that tilted about a horizontal axis in a northwest direction.

The following scenario can be proposed for the history of the Karaginsky accretionary prism.

The accretionary structures of Karaginsky Island and possibly of East Kamchatka may have begun to form in Early–Middle Eocene by the tectonic juxtaposition of the ophiolitic and Late Cretaceous–Paleocene island-arc units, the latter being devoid of ophiolitoclastics (Kravchenko-Berezhnoy & Nazimova 1991). Prior to that time, the East Kamchatka and Olyutorsky island arcs, whose remnants are found in the accretionary prism, migrated from latitudes of about 40–55°N in a roughly north to north-northwest direction (Kovalenko 1992a,b; Chekhovich *et al.* 1990; Engebretson *et al.* 1985) together with the Kula plate, the Benioff zone beneath these arcs probably plunging to the south (Kovalenko 1992; Chekhovich *et al.* 1990; Kravchenko-Berezhnoy 1989) (Fig. 10a). The fact that the island-arc sequences show distinct deformation ages in South Koryakia (Miocene) (Chekhovich *et al.* 1990) and East Kamchatka (Eocene) (Zinkevich & Tsukanov 1992) provides grounds to believe that the drifting arcs

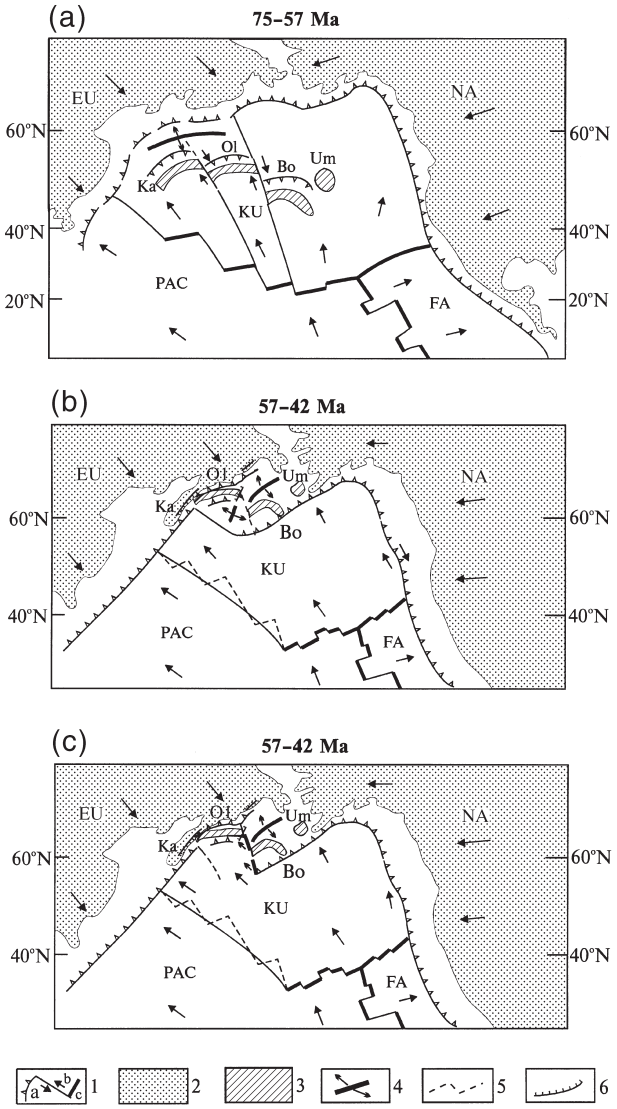


Fig. 10 Paleoreconstructions for (a) Campanian–Maastrichtian time and (b, c) Eocene time (b, assuming that the Komandorsky shear was incepted at the Paleocene–Eocene transition; c, assuming that the shear was incepted not earlier than Late Eocene (42 Ma)). 1, plate boundaries (a, subduction zones; b, transform faults; c, spreading ridges); 2, continents; 3, terranes; 4, spreading zones; 5, ancient plate boundaries; 6, fold-and-thrust belts. Arrows indicate direction of plate motion. EU, Eurasia; FA, Farallon plate; KU, Kula plate; NA, North America; PAC, Pacific plate. Terranes: Bo, Bowers; Ka, Kamchatka; Ol, Olyutorsky; Um, Umnak. The pacific plate kinematics are taken from Londsale (1988) and Engebretson *et al.* (1985).

were segmented by transform faults, the resulting segments colliding with the continental margin at different times. A transform boundary may have separated the Bowers block, now occurring to the east of the Olyutorsky island arc (Fig. 10a).

At the Paleocene/Eocene transition or in the Early Eocene time, a rearrangement of plate motions took place in the Pacific. Spreading direction in the Kula Ridge switched from north–south

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to northwest–southeast (Lonsdale 1988). At about the same time, the East Kamchatka island arc and possibly the Umnak Plateau collided with a continent, and at least the eastern part of the Aleutian arc was incepted. This arc extends, according to different workers, as far as 175°E. It may have been due to this event that subduction polarity in the Kamchatka–Olyutorsky island-arc system changed from south-southeast to northwest, and the Karaginsky–East Kamchatka accretionary prism began to form. This is evidenced by the appearance of ophiolitic and arc-derived clastics in the Middle–Upper Eocene flysch sequences on Karaginsky Island and Ozernoi Peninsula.

The fact that the northwest-directed subduction at the Karaginsky accretionary prism (marked by the arc-related volcanic–terrigenous complex) is devoid of strike-slip component may have been due to either sea-floor spreading in the Komandorsky Basin (Fig. 10b) (as suggested by the presence of MORB-like tholeiites in the flysch of Karaginsky Island and of Kumroch Range in Kamchatka) or to underthrusting of one of the Pacific plates (Fig. 10c). The latter option stipulates that the Komandorsky shear was non-existent, and the transform fault that separated the Olyutorsky island arc and the Bowers block evolved into a transtensional structure (Fig. 10c). In this situation, sea-floor spreading supposedly taking place in this structure would account for the MORB-like basalts present in the Eocene flysch on Karaginsky Island (Kravchenko-Berezhnoy *et al.* 1990) and, possibly, in the Kumroch Range (Tsukanov & Fedorchuk 1991).

At 42 Ma, the Komandorsky shear was incepted (Engebretson *et al.* 1985). Hence, the development of the Karaginsky accretionary prism may have been related only to backarc spreading in the Komandorsky Basin, which is believed to continue today. Prior to the second deformation event (which is recorded paleomagnetically), the volcanic–terrigenous and flysch sequences dipped steeply (50–80°) to the south-southeast; this dip might have been acquired through the Late Eocene stage of inception of the accretionary prism. The subsequent tilting of the volcanic–terrigenous and flysch complexes may have occurred in post-Early Oligocene time (30 Ma; the upper age limit for the flysch complex) but most probably predated the overlapping lower Miocene molasse complex. This tilting must have been related to the late stage of back-arc spreading in the Komandorsky Basin.

CONCLUSIONS

The NRM of rocks on the Karaginsky Island yield both prefolding (close to the original) and post-folding components.

Vectors of the prefolding NRM components suggest that the Maastrichitan–Paleocene island-arc sequences of Karaginsky Island (volcanic–terrigenous complex) were deposited at latitudes of about 40–55°N and originally did not form part of either Eurasia ($F = 19$, $\Delta F = 6.5$) or North America ($F = 17$, $\Delta F = 4.4$).

The studied blocks rotated by 280 ± 4.40 counterclockwise in the horizontal relative to the poleward direction, insignificantly ($R = -15$, $\Delta R = 40$) relative to Eurasia, and by 440 ± 170 ($R = 44$, $\Delta R = 17$) counterclockwise with respect to North America. This implies that the Karaginsky accretionary prism arose without any significant strike-slip motions, and plate convergence at this prism must have been nearly head-on.

Similar latitudes of formation and senses of rotation obtained from coeval deposits in the northeastern Olyutorsky zone (Kovalenko 1992a,b; Kovalenko 1996) and in East Kamchatka (Levashova *et al.* 1999) suggest that the Karaginsky Maastrichtian–Paleocene island-arc sequence originally formed part of the Olyutorsky island arc in South Koryakia.

Analysis of secondary NRM components reveals that the Karaginsky structure experienced at least two stages of deformation. It may have been during the first stage (possibly Late Eocene) that the sequences were tilted 50–80° to the south-southwest and during the second stage (Late Oligocene?) that they were tilted by 45–65° in west-northwest to northwest directions. This coincides with the northwest vergence of the present-day structure of Karaginsky Island.

ACKNOWLEDGEMENTS

We are grateful to N. A. Bogdanov, S. D. Sokolov and V. D. Chekhovich for helpful discussion.

This work was supported by the Russian Foundation for Fundamental Research, project nos 97-05-64609 and 97-05-64610.

REFERENCES

BAZHENOV M. L. & SHIPUNOV S. V. 1988. [Fold Test in Paleomagnetism.] *Izvestiya, Physics of the Solid Earth* **7**, 89–101. (In Russian.)

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

BECK M. E., Jr. 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. *Journal of Geophysical Research* **85**, 7115–31.

BESSE J. & COURTILLOT V. 1991. Revised and synthetic polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma. *Journal of Geophysical Research* **96**, 4029–50.

BOGDANOV N. A., CHEKHOVICH V. D., SUKHOV A. N. & VISHNEVSKAYA V. S. 1982. Tectonics of the Olutorsky zone. *Tectonic Essays of the Koryak Highlands*, pp. 189–217. Nauka Publishing, Moscow.

CHEKHOVICH V. D., BOGDANOV N. A., KRAVCHENKO-BEREZHNOY I. R., AVERINA G. YU., GLADENKOV A. YU. & TIL'MAN S. M. 1990. [*Geology of the western Bering Sea.*] ILSAN, Moscow. (In Russian.)

DEMAREST H. H., Jr. 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. *Journal of Geophysical Research* **88**, 4121–328.

ENGEBRETSON D. S., COX A. & GORDON R. G. 1985. Relative motion between oceanic and continental plates in the Pacific basin. *Geological Society of America, Special Paper* **206**.

GEIST E. I., VALLIER T. L. and SCHOLL D. W. 1994. Origin, transport, and emplacement of an exotic island-arc terrane exposed in Eastern Kamchatka, Russia. *Geological Society of America Bulletin* **106**, 1182–94.

GLADENKOV YU. B. 1972. Nauka Publishing, Neogene Deposits of Kamchatka. Moscow.

KHRAMOV A. N., GONCHAROV G. I., KOMISSAROVA R. A. *et al.* 1982. *Paleomagnetology*. Nedra, Leningrad.

KIRSCHVINK J. L. 1980. The least-squares line and plane and the analysis of Paleomagnetic Data. *Geophysical Journal of Royal Astronomy Society* **62**, 699–718.

KOVALENKO D. V. 1992a. [Paleomagnetism of the Paleogene complexes of Il'pinsky Peninsula.] *Geotectonics* **5**, 78–95. (In Russian.)

KOVALENKO D. V. 1992b. [*Geological interpretation of paleomagnetic data from Late Cretaceous and Paleogene complexes of the Olutorsky zone*], Author's abstract of candidate's dissertation. ILSAN, Moscow. (In Russian.)

KOVALENKO D. V. 1996. [Paleomagnetism and kinematics of the central part of the Olutorsky Range (Koryak Highlands)] *Geotectonics* **3**, 82–96. (In Russian.)

KRAVCHENKO-BEREZHNOY I. R. 1989. [*Geologic Setting of Magmatic Complexes in the Western Rim of the Komandorsky Basin*], Author's abstract of candidate's dissertation. ILSAN, Moscow. (In Russian.)

KRAVCHENKO-BEREZHNOY I. R. & NAZIMOVA YU. V. 1991. The Cretaceous Ophiolite of Karaginsky Island (the Western Bering Sea). *Ofioliti* **16**, 79–110.

KRAVCHENKO-BEREZHNOY I. R., CHAMOV N. P. and SHCHERBININA E. A. 1990. MORB-like tholeiites in a Late Eocene turbidite sequence on Karaginsky Island (the Western Bering Sea). *Ofioliti* **15**, 231–50.

LEVASHOVA N. M., BAZHENOV M. L. and SHAPIRO M. N. 1999. Late Cretaceous paleomagnetism of the east range island-arc complex, Kamchatka: Implications for terrane movements and kinematics of the North-West Pacific. *Tectonophysics* (in press).

LONSDALE P. 1988. Paleogene history of the Kula Plate: Offshore evidence and onshore implications. *Geological Society of America Bulletin* **100**, 733–54.

MCFADDEN P. L. and JONES D. L. 1981. The fold test in paleomagnetism. *Geophysical Journal of Royal Astronomy Society* **67**, 53–8.

SHAPIRO M. N. and PETRINA N. K. 1985. New data on the geologic structure of Karaginsky Island. *Pacific Geology* **1**, 27–33.

TSUKANOV N. V. and FEDORCHUK A. V. 1991. Oceanic complexes in the structure of the Eastern Kumroch Range (Kamchatka). *Doklady Earth Sciences* **307**, 943–8.

ZIJDERVELD J. D. A. 1967. Demagnetization in rocks: Analysis of results, methods in paleomagnetism, *In* Collinson *et al.* eds. pp. 254–86. Elsevier, New York.

ZINKEVICH V. P. and TSUKANOV N. V. 1992. Formation of accretionary structure of East Kamchatka in Late Mesozoic–Early Cenozoic time. *Geotectonics* **4**, 97–112.

14401738, 1999, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1440-1738.1999.00243.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License